STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 10 -

Assist. Prof. Dr. Muzaffer BÖREKÇİ
Email: structuralanalysis1mb@gmail.com

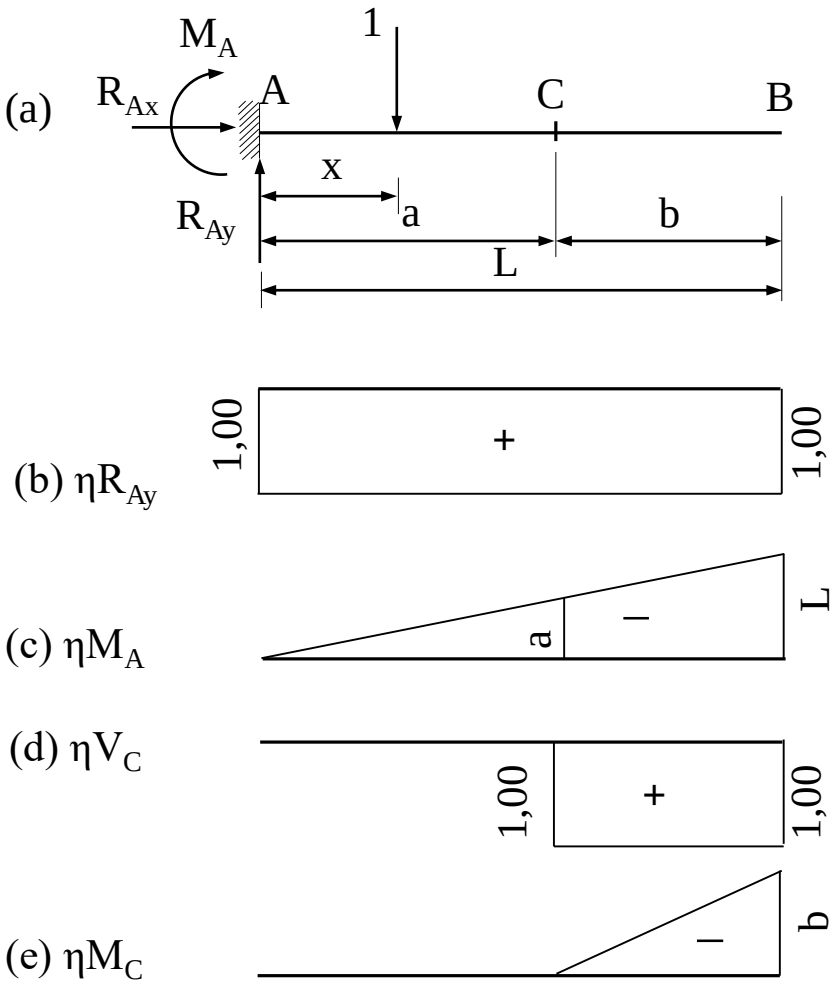
Influence Lines of Statically Determinate Systems

Simply Supported Beam

Covered while explaining the analytical method.

Cantilever Beam

(i) Unit vertical load:



Support Reactions:

- $\Sigma P_x = 0 \rightarrow R_{Ax} = 0 \; ; \; \eta R_{Ax} = 0$
- $\Sigma P_y = 0 \rightarrow R_{Ay} = 1 \; ; \; \eta R_{Ay} = 1$
- $\Sigma M_A = 0 \rightarrow M_A = -x \; ; \; \eta M_A = -x$

Shear Force and Bending Moment at C

$0<x<a$

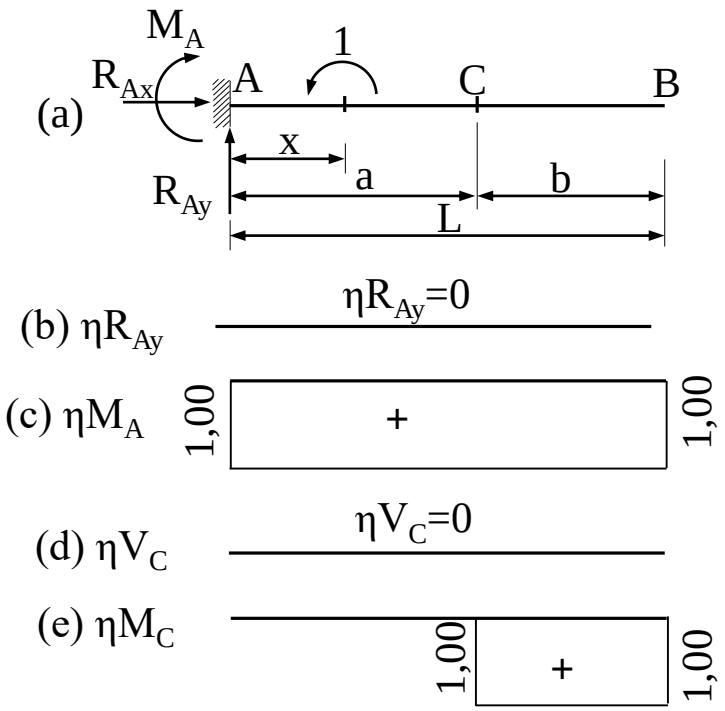
- $\Sigma P_y = 0 \rightarrow V_C = 0 \; ; \; \eta V_C = 0$
- $\Sigma M_C = 0 \rightarrow M_C = 0 \; ; \; \eta M_C = 0$

$a<x<L$

- $\Sigma P_y = 0 \rightarrow V_C = 1 \; ; \; \eta V_C = 1$
- $\Sigma M_C = 0 \rightarrow M_C = -(x-a) \; ; \; \eta M_C = -(x-a)$

Cantilever Beam

(ii) Unit moment:



Support Reactions:

- $\Sigma P_x = 0 \rightarrow R_{Ax} = 0 \; ; \; \eta R_{Ax} = 0$
- $\Sigma P_y = 0 \rightarrow R_{Ay} = 0 \; ; \; \eta R_{Ay} = 0$
- $\Sigma M_A = 0 \rightarrow M_A = 1 \; ; \; \eta M_A = 1$

Shear Force and Bending Moment at C

$0<x<a$

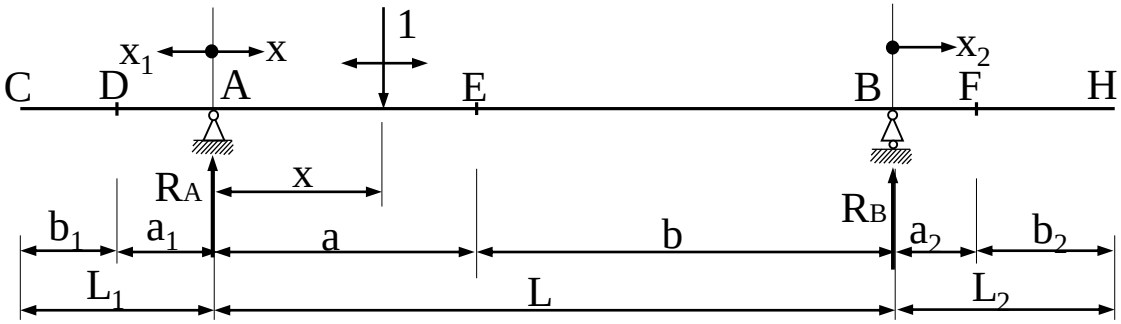
- $\Sigma P_y = 0 \rightarrow V_C = 0 \; ; \; \eta V_C = 0$
- $\Sigma M_C = 0 \rightarrow M_C = 0 \; ; \; \eta M_C = 0$

$a<x<L$

- $\Sigma P_y = 0 \rightarrow V_C = 0 \; ; \; \eta V_C = 0$
- $\Sigma M_C = 0 \rightarrow M_C = 1 \; ; \; \eta M_C = 1$

Overhanging Beam

(i) Unit vertical load:



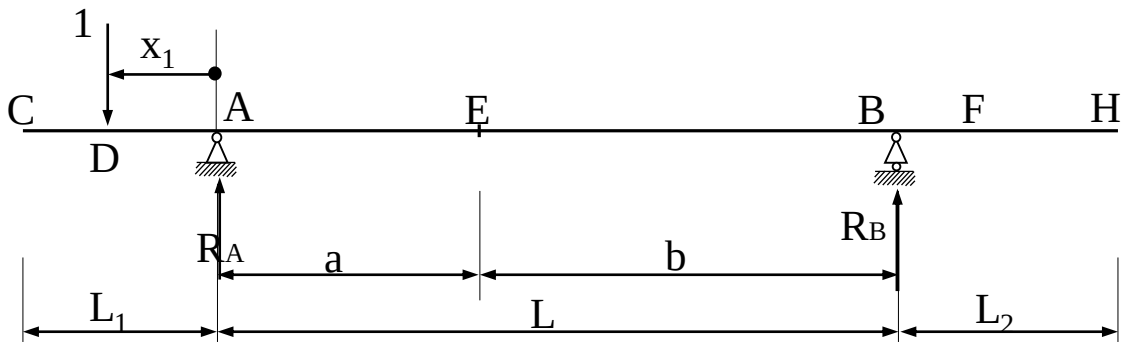
Unit load is in between A and C ($0<x_1<L_1$)

Support Reactions:

- $\Sigma M_B = 0 \rightarrow R_{Ay} = (L+x_1)/L$; $\eta R_{Ay} = (L+x_1)/L$
- $\Sigma M_A = 0 \rightarrow R_{By} = -x_1/L$; $\eta R_{By} = -x_1/L$

$x_1 = 0 \rightarrow$ **$\eta R_{Ay} = 1$; $\eta R_{By} = 0$**

$x_1 = L_1 \rightarrow$ **$\eta R_{Ay} = 1+L_1/L$; $\eta R_{By} = -L_1/L$**



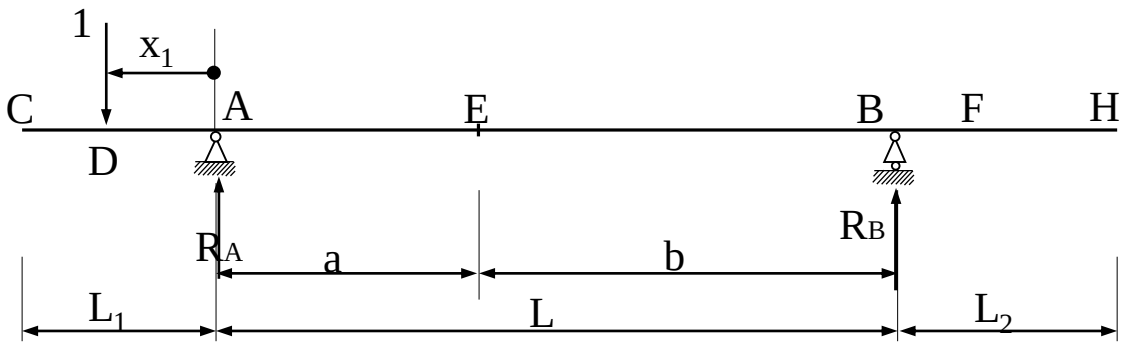
Unit load is in between A and C ($0<x_1<L_1$)

Shear force and bending moment on the left of A:

- $V_{AD} = -1$; $\eta V_{AD} = -1$
- $M_{AD} = -x_1$; $\eta M_{AD} = -x_1$

$x_1 = 0 \rightarrow \mathbf{\eta M_{AD} = 0}$

$x_1 = L_1 \rightarrow \mathbf{\eta M_{AD} = -L_1}$

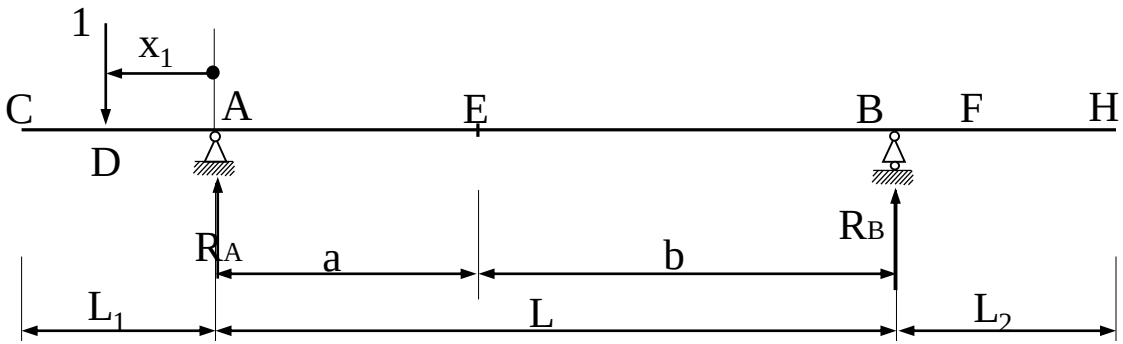


Unit load is in between A and C ($0<x_1<L_1$)

Shear force and bending moment at D:

$0<x_1<a_1$

- $V_D = 0$; $\eta V_D = 0$
- $M_D = 0$; $\eta M_D = 0$



$a_1<x_1<L_1$

- $V_D = -1$; $\eta V_D = -1$
- $M_D = -(x_1 - a_1)$; $\eta M_D = -(x_1 - a_1)$

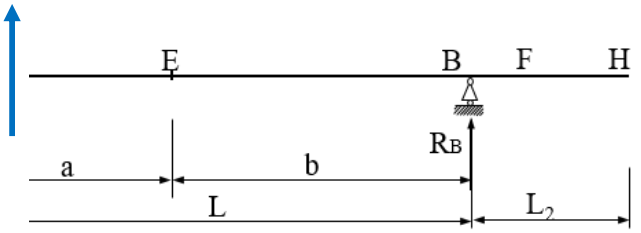
$x_1 = a_1 \rightarrow \mathbf{\eta M_D = 0}$

$x_1 = L_1 \rightarrow \mathbf{\eta M_D = -b_1}$

Unit load is in between A and C ($0<x_1<L_1$)

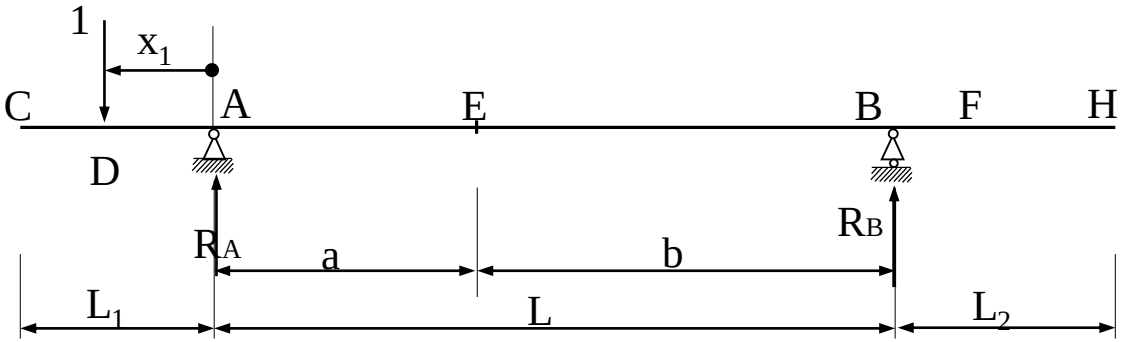
Shear force on the right of A:

- $V_{AE} = -R_{By}$; $\eta V_{AE} = -\eta R_{By} = x_1/L$



$x_1 = 0 \rightarrow \mathbf{\eta V_{AE} = 0}$

$x_1 = L_1 \rightarrow \mathbf{\eta V_{AE} = L_1/L}$



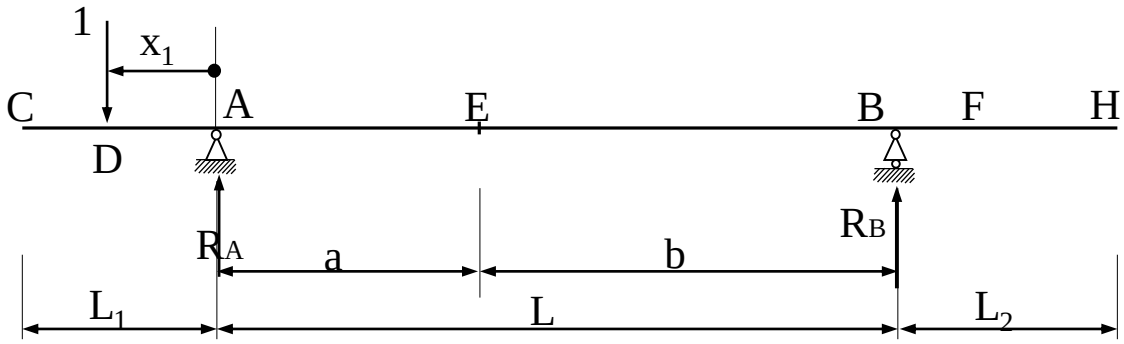
Unit load is in between A and C ($0<x_1<L_1$)

Shear force and bending moment at E:

- $V_E = -R_{By}$; $\eta V_E = -\eta R_{By} = x_1/L$
- $M_E = b.R_{By}$; $\eta M_E = b.\eta R_{By} = -b.x_1/L$

$x_1 = 0 \rightarrow$ **$\eta V_E = 0$; $\eta M_E = 0$**

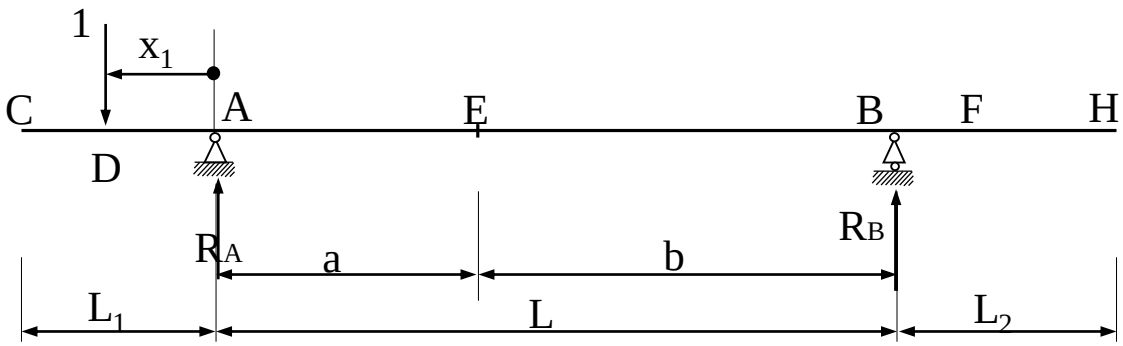
$x_1 = L_1 \rightarrow$ **$\eta V_E = L_1/L$; $\eta M_E = -b.L_1/L$**



Unit load is in between A and C ($0<x_1<L_1$)

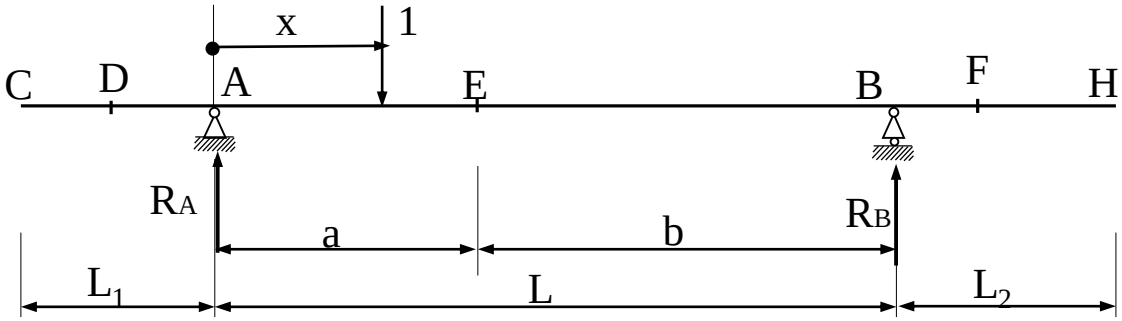
Shear force and bending moment at B and F:

- $M_{BF} = M_F = V_{BF} = V_F = 0$



Unit load is in between A and B (0<x<L)

For this situation, influence lines are same with the simply supported beam. When the load is in between A and B, no internal force occurs in the overhanging parts.



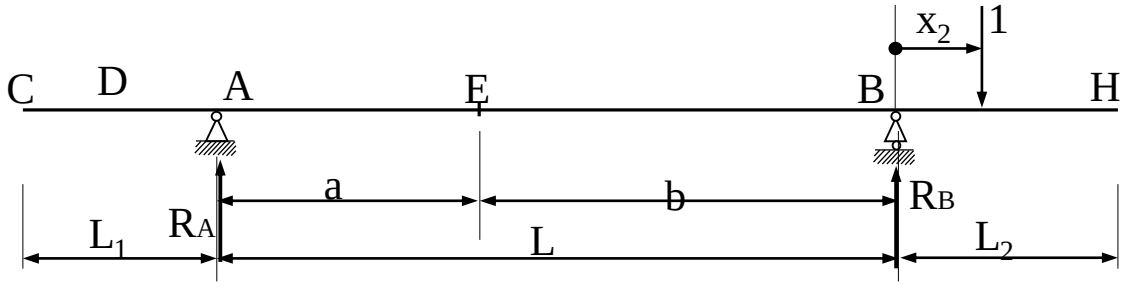
Unit load is in between B and H ($0<x_2<L_2$)

Support Reactions:

- $\Sigma M_B = 0 \rightarrow R_{Ay} = -x_2/L$; $\eta R_{Ay} = -x_2/L$
- $\Sigma M_A = 0 \rightarrow R_{By} = (L+x_2)/L$; $\eta R_{By} = (L+x_2)/L$

$x_2 = 0 \rightarrow$ **$\eta R_{Ay} = 0$; $\eta R_{By} = 1$**

$x_2 = L_2 \rightarrow$ **$\eta R_{Ay} = -L_2/L$; $\eta R_{By} = 1+L_2/L$**



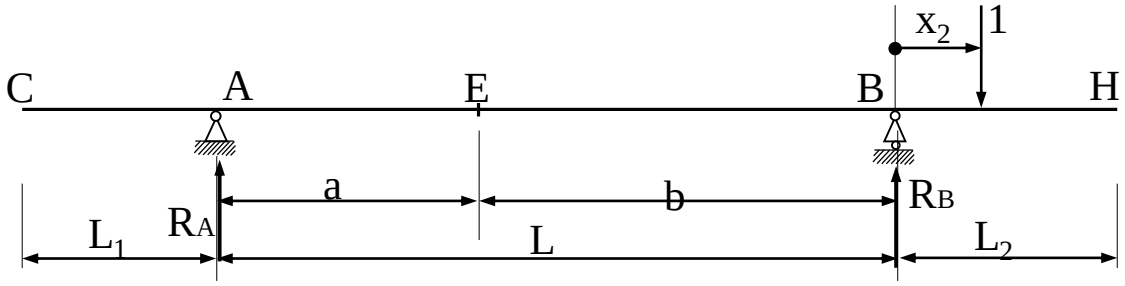
Unit load is in between B and H ($0<x_2<L_2$)

Shear force on the right of A:

- $V_{AE} = R_{Ay}$; $\eta V_{AE} = \eta R_{Ay} = -x_2/L$

$x_2 = 0 \rightarrow \mathbf{\eta V_{AE} = 0}$

$x_2 = L_2 \rightarrow \mathbf{\eta V_{AE} = -L_2/L}$



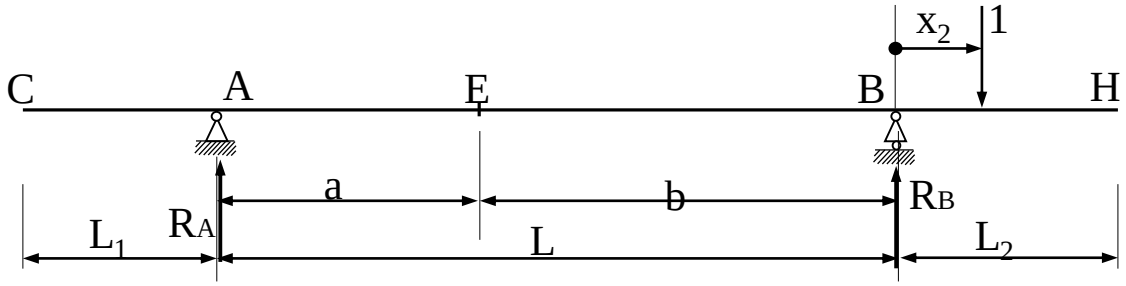
Unit load is in between B and H ($0<x_2<L_2$)

Shear force and bending moment at E:

- $V_E = R_{Ay}$; $\eta V_E = \eta R_{Ay} = -x_2/L$
- $M_E = a.R_{Ay}$; $\eta M_E = a.\eta R_{Ay} = -a.x_2/L$

$x_2 = 0 \rightarrow \mathbf{\eta V_E = 0 \ ; \eta M_E = 0}$

$x_2 = L_2 \rightarrow \mathbf{\eta V_E = -L_2/L \ ; \eta M_E = -a.L_2/L}$



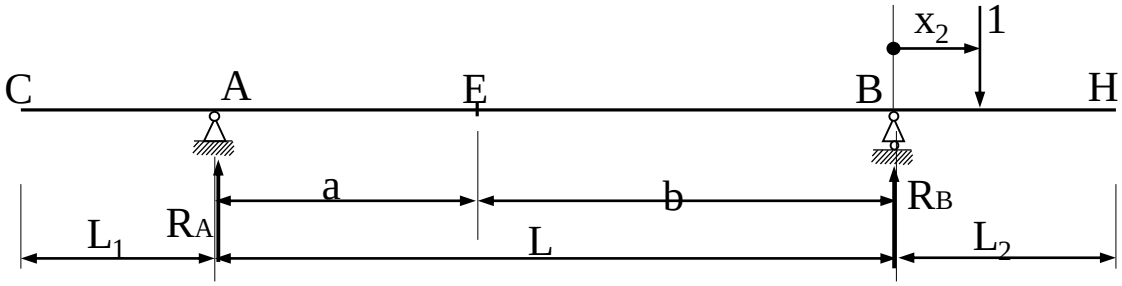
Unit load is in between B and H ($0<x_2<L_2$)

Shear force and bending moment on the right of B:

- $V_{BF} = 1$; $\eta V_{BF} = 1$
- $M_{BF} = -x_2$; $\eta M_{BF} = -x_2$

$x_2 = 0 \rightarrow$ **$\eta V_{BF} = 1$; $\eta M_{BF} = 0$**

$x_2 = L_2 \rightarrow$ **$\eta V_{BF} = 1$; $\eta M_{BF} = -L_2$**

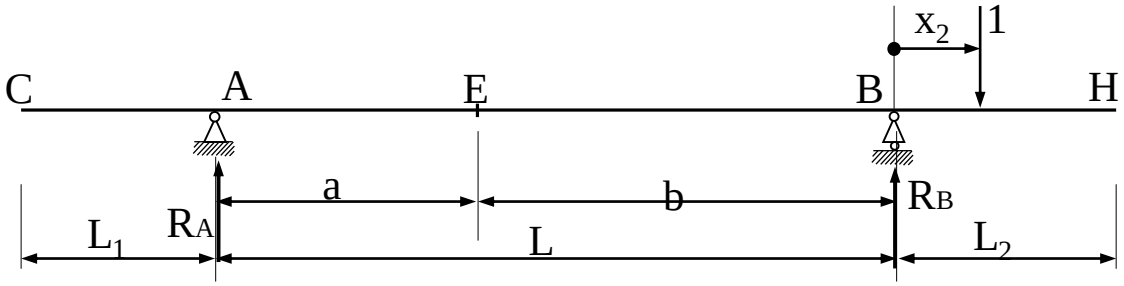


Unit load is in between B and H ($0<x_2<L_2$)

Shear force and bending moment at F:

$0<x_2<a_2$

- $V_F = 0$; $\eta V_F = 0$
- $M_F = 0$; $\eta M_F = 0$



$a_2<x_2<L_2$

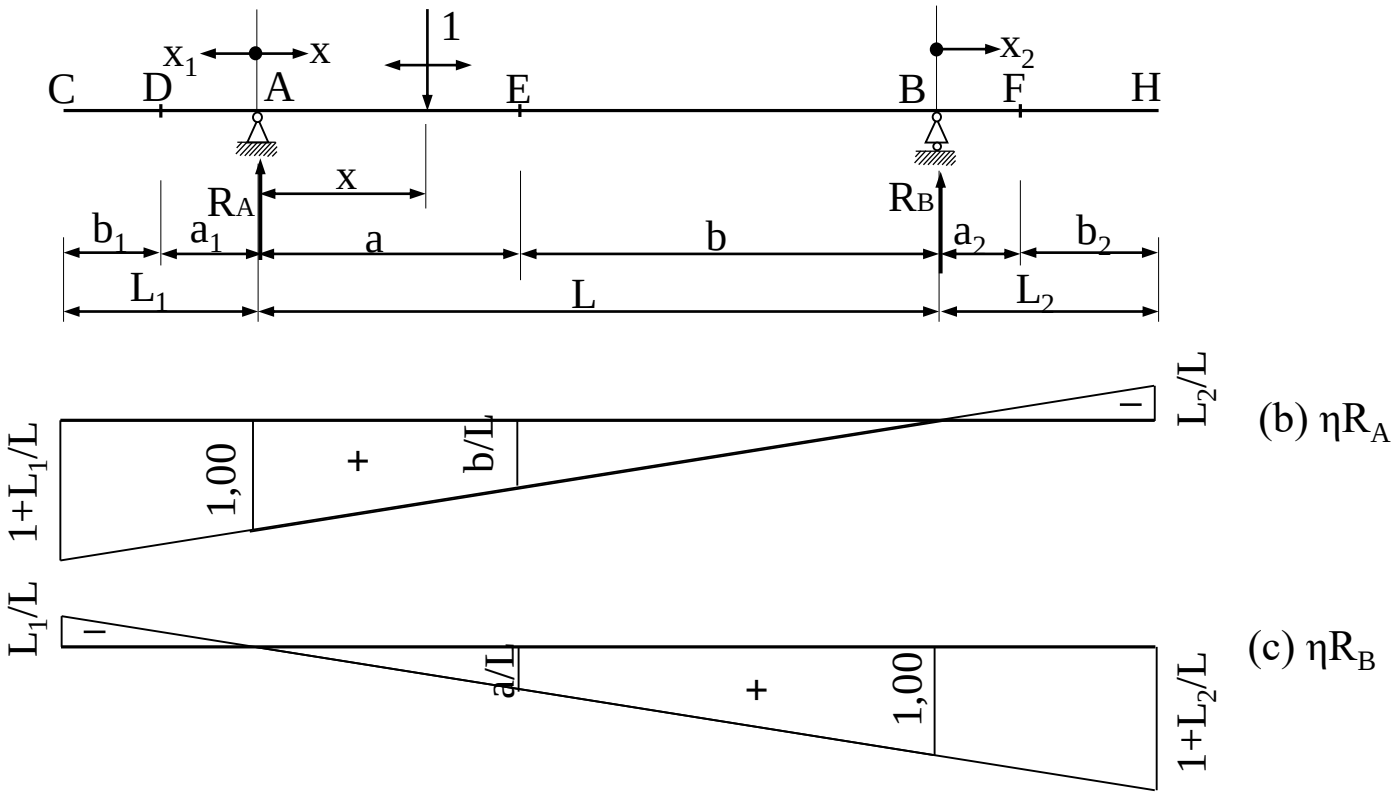
- $V_F = 1$; $\eta V_F = 1$
- $M_F = -(x_2 - a_2)$; $\eta M_F = -(x_2 - a_2)$

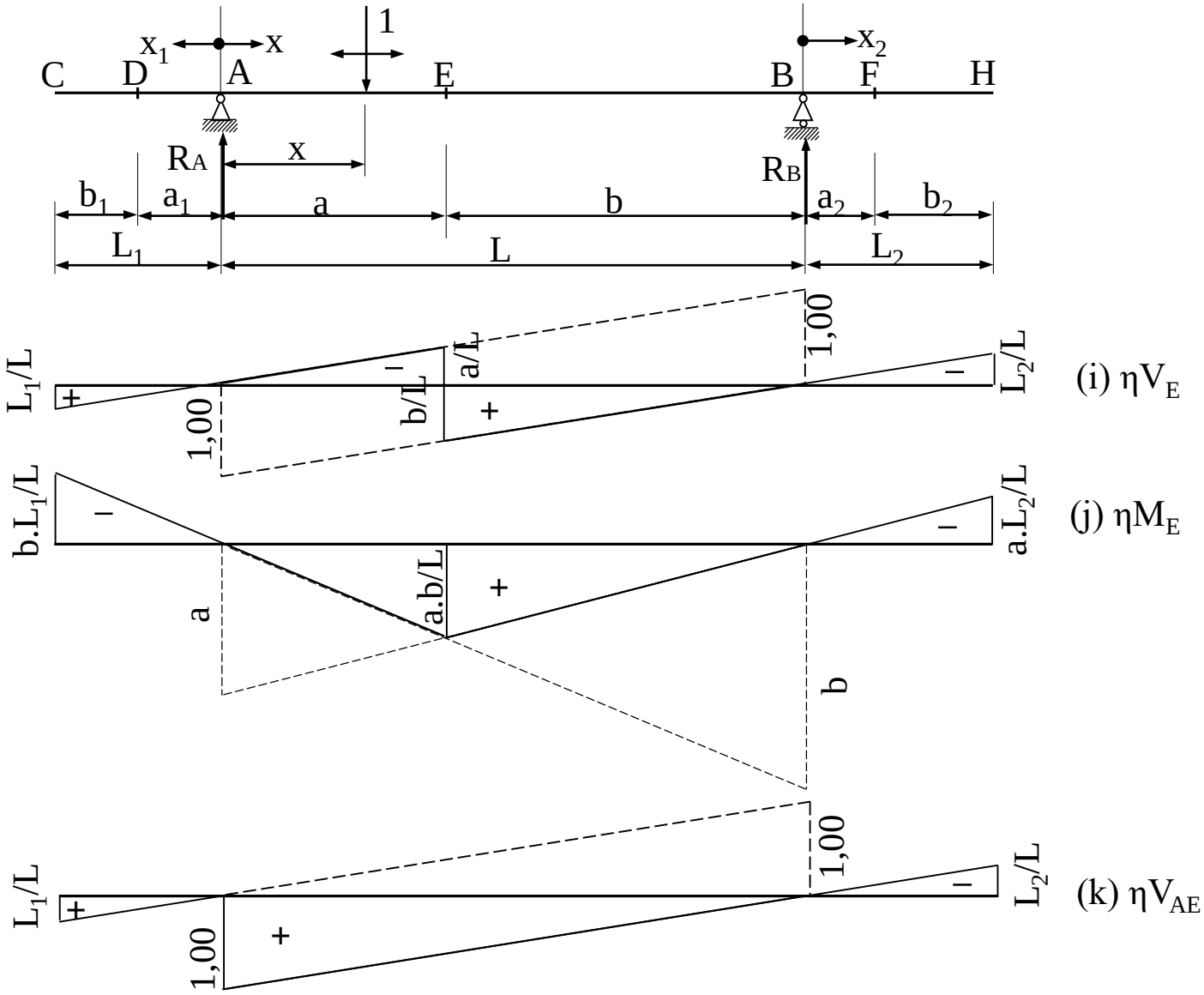
$x_2 = a_2 \rightarrow \mathbf{\eta M_F = 0}$

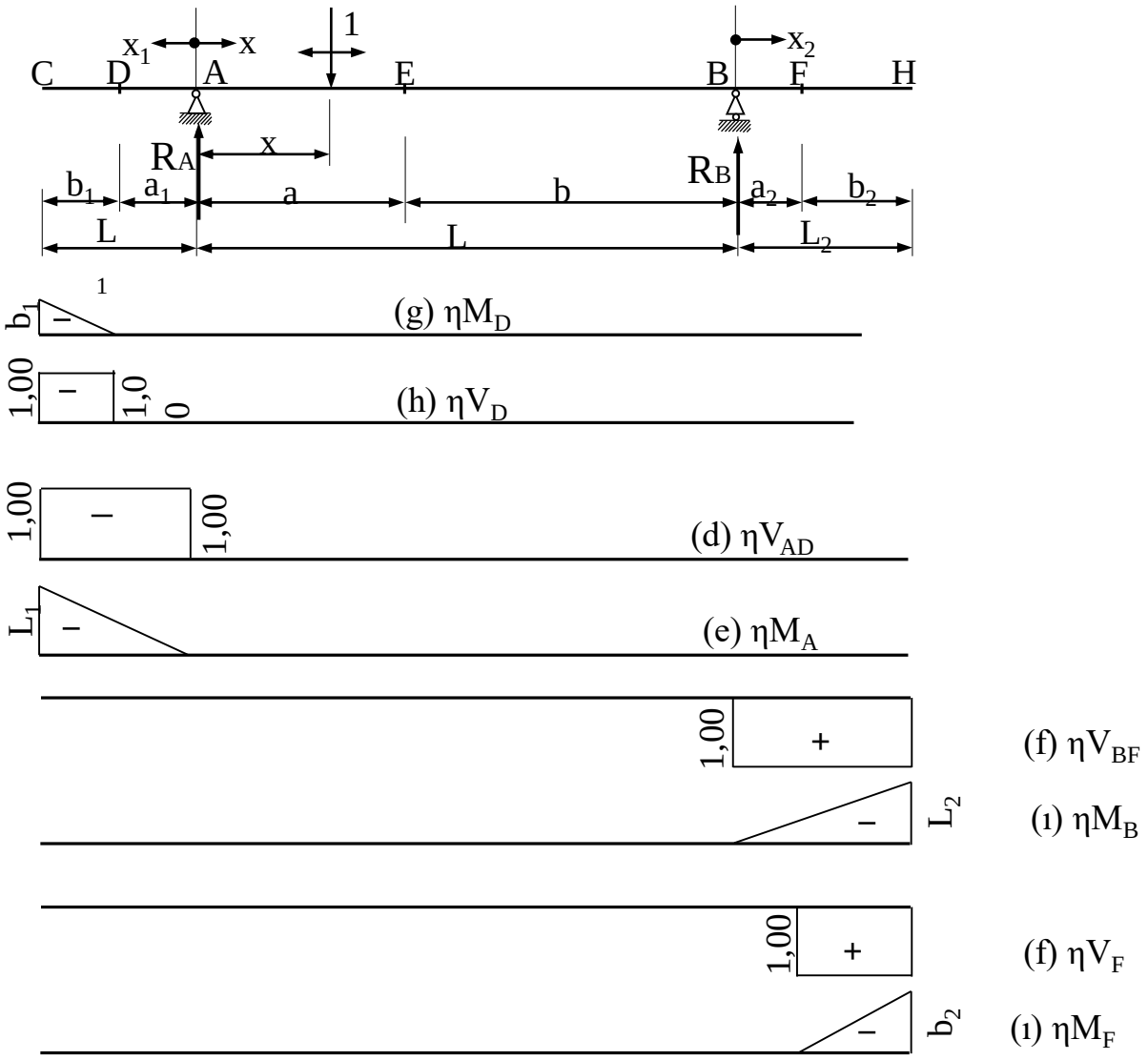
$x_2 = L_2 \rightarrow \mathbf{\eta M_F = -b_2}$

NOTE1: Influence lines of shear force and bending moment for overhanging parts are the same as those for a cantilever beam.

NOTE2: Influence lines of support reactions, shear force and bending moment are the same as those for a simply supported beam.

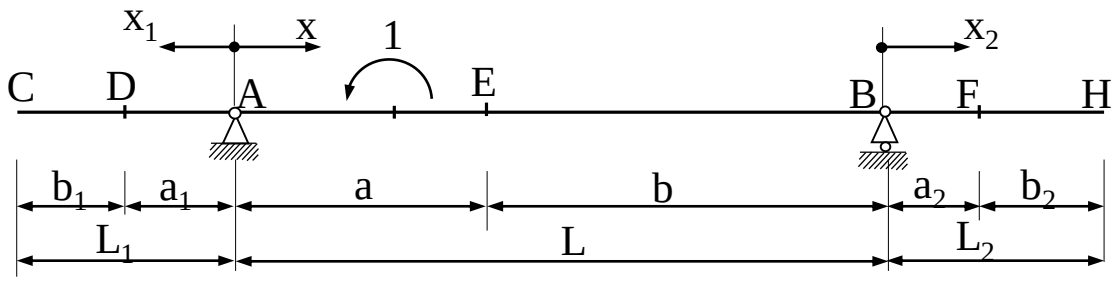

C
D
A
E
B
F
H
x1
x
1
x2
RA
RB
x
b1
a1
a
b
a2
b2
L1
L
L2
1+L1/L
1,00
+
b/L
L2/L
(b) ηRA
L1/L
a/L
+
1,00
1+L2/L
(c) ηRB


x1
x
1
x2
C
D
A
E
B
F
H
RA
x
RB
b1
a1
a
b
a2
b2
L1
L
L2
L1/L
1,00
a/L
b/L
L2/L
(i) ηVE
b.L1/L
a.L2/L
a
a.b/L
b
(j) ηME
L1/L
1,00
L2/L
(k) ηVAE


x1
x
1
x2
C
D
A
E
B
F
H
RA
x
RB
b1
a1
a
b
a2
b2
L
L
L2
1
b1
(g) ηMD
1,00
1,0
0
(h) ηVD
1,00
1,00
(d) ηVAD
L1
(e) ηMA
1,00
(f) ηVBF
L2
(ı) ηMB
1,00
(f) ηVF
b2
(ı) ηMF

Overhanging Beam

(i) Unit moment:



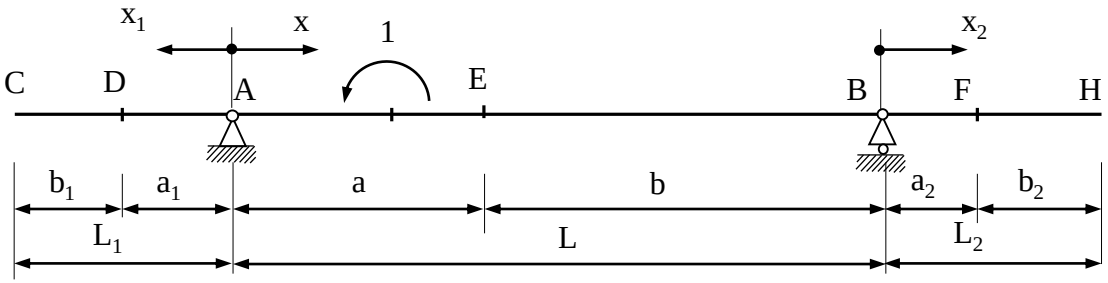
Unit moment is in between A and C ($0<x_1<L_1$)

Support Reactions:

- $\Sigma M_B = 0 \rightarrow R_{Ay} = 1/L$; $\eta R_{Ay} = 1/L$
- $\Sigma M_A = 0 \rightarrow R_{By} = -1/L$; $\eta R_{By} = -1/L$

Shear force and bending moment on the left of A:

- $V_{AD} = 0$; $\eta V_{AD} = 0$
- $M_{AD} = -1$; $\eta M_{AD} = -1$



Unit moment is in between A and C ($0<x_1<L_1$)

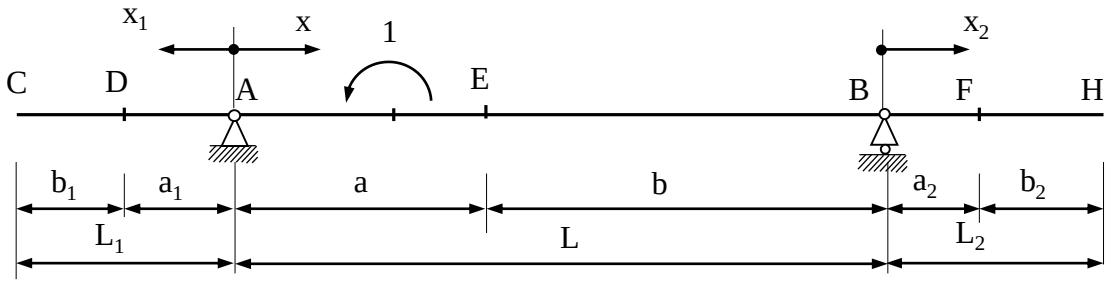
Shear force and bending moment at D:

$0<x_1<a_1$

- $V_D = 0$; $\eta V_D = 0$
- $M_D = 0$; $\eta M_D = 0$

$a_1<x_1<L_1$

- $V_D = 0$; $\eta V_D = 0$
- $M_D = -1$; $\eta M_D = -1$



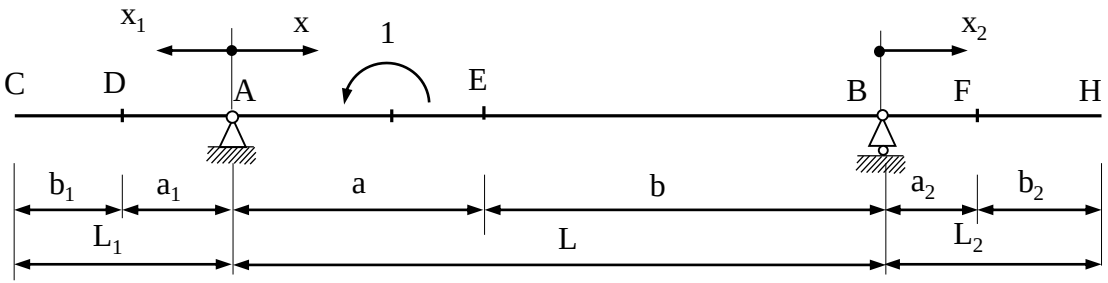
Unit moment is in between A and C ($0<x_1<L_1$)

Shear force on the right of A:

- $V_{AE} = -R_{By}$; $\eta V_{AE} = -\eta R_{By} = 1/L$

Shear force and bending moment at E:

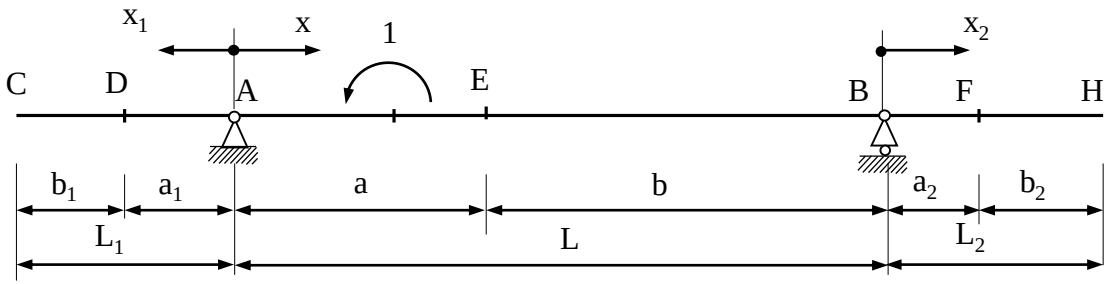
- $V_E = -R_{By}$; $\eta V_E = -\eta R_{By} = 1/L$
- $M_E = b.R_{By}$; $\eta M_E = b.\eta R_{By} = -b/L$



Unit moment is in between A and C ($0<x_1<L_1$)

Shear force and bending moment at B and F:

- $M_{BF} = M_F = V_{BF} = V_F = 0$



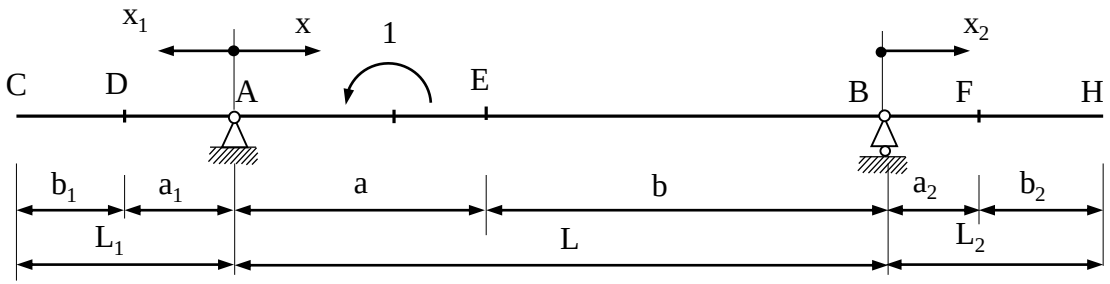
Unit moment is in between A and B (0<x<L)

Support Reactions:

- $\Sigma M_B = 0 \rightarrow R_{Ay} = 1/L$; $\eta R_{Ay} = 1/L$
- $\Sigma M_A = 0 \rightarrow R_{By} = -1/L$; $\eta R_{By} = -1/L$

Since the load is in between A and B, no internal force occurs in the overhanging parts

- $M_{AD} = M_D = M_{BF} = M_F = V_{AD} = V_{BF} = V_D = V_F = 0$



Unit moment is in between A and B (0<x<L)

Shear force on the right of A:

- $V_{AE} = R_{Ay}$; $\eta V_{AE} = \eta R_{Ay} = 1/L$

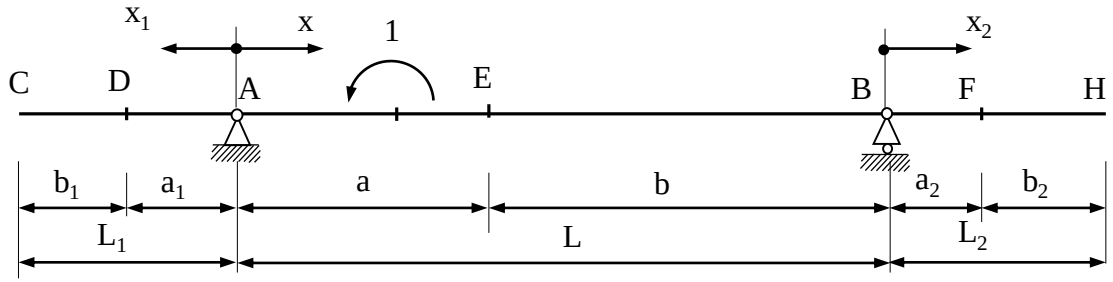
Shear force and bending moment at E:

0<x<a

- $V_E = -R_{By}$; $\eta V_E = -\eta R_{By} = 1/L$
- $M_E = bR_{By}$; $\eta M_E = b\eta R_{By} = -b/L$

a<x<L

- $V_E = R_{Ay}$; $\eta V_E = \eta R_{Ay} = 1/L$
- $M_E = aR_{Ay}$; $\eta M_E = a\eta R_{Ay} = a/L$



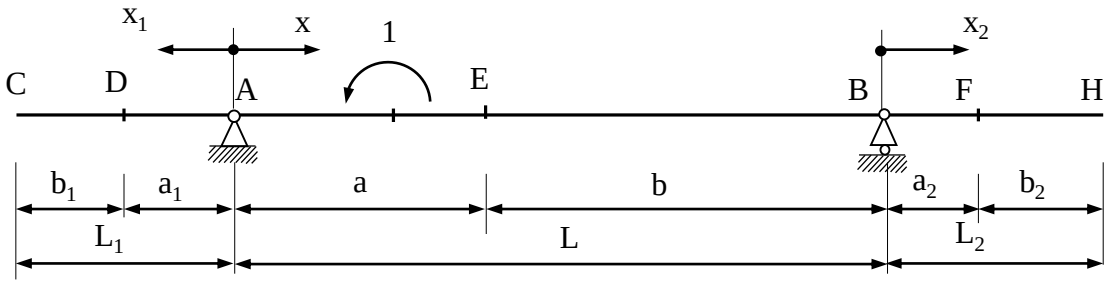
Unit moment is in between B and H ($0<x_2<L_2$)

Support Reactions:

- $\Sigma M_B = 0 \rightarrow R_{Ay} = 1/L$; $\eta R_{Ay} = 1/L$
- $\Sigma M_A = 0 \rightarrow R_{By} = -1/L$; $\eta R_{By} = -1/L$

Since there is no load acting on the other overhanging part, no internal force occurs in that part.

- $M_{AD} = M_D = V_{AD} = V_D = 0$



Unit moment is in between B and H ($0<x_2<L_2$)

Shear force on the right of A:

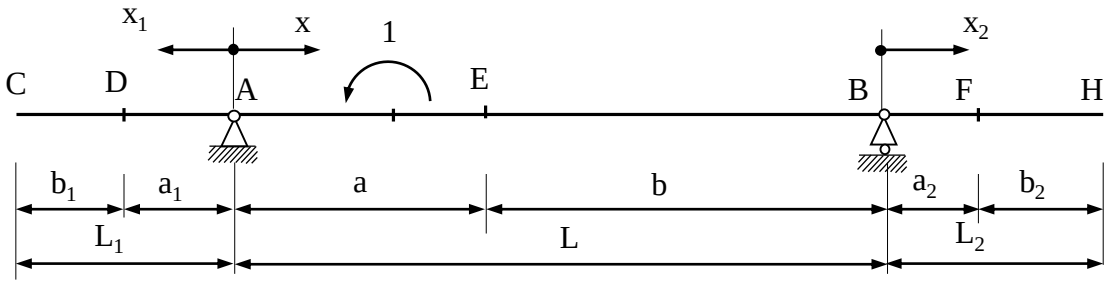
- $V_{AE} = R_{Ay}$; $\eta V_{AE} = \eta R_{Ay} = 1/L$

Shear force and bending moment at E:

- $V_E = R_{Ay}$; $\eta V_E = \eta R_{Ay} = 1/L$
- $M_E = a.R_{Ay}$; $\eta M_E = a.\eta R_{Ay} = a/L$

Shear force and bending moment on the right of B:

- $V_{BF} = 0$; $\eta V_{BF} = 0$
- $M_{BF} = 1$; $\eta M_{BF} = 1$



Unit moment is in between B and H ($0<x_2<L_2$)

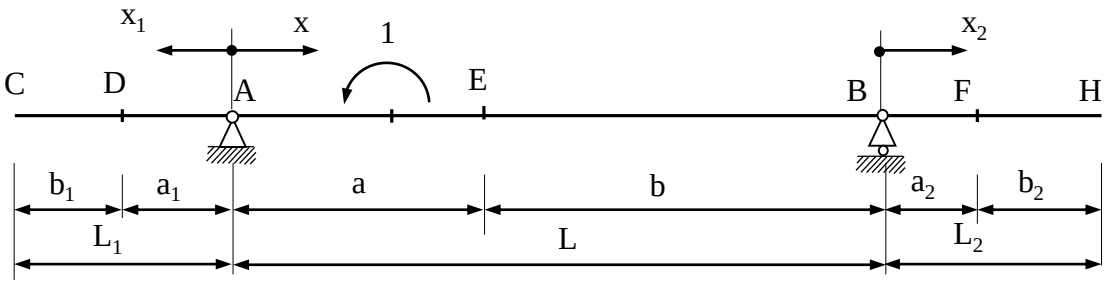
Shear force and bending moment at F:

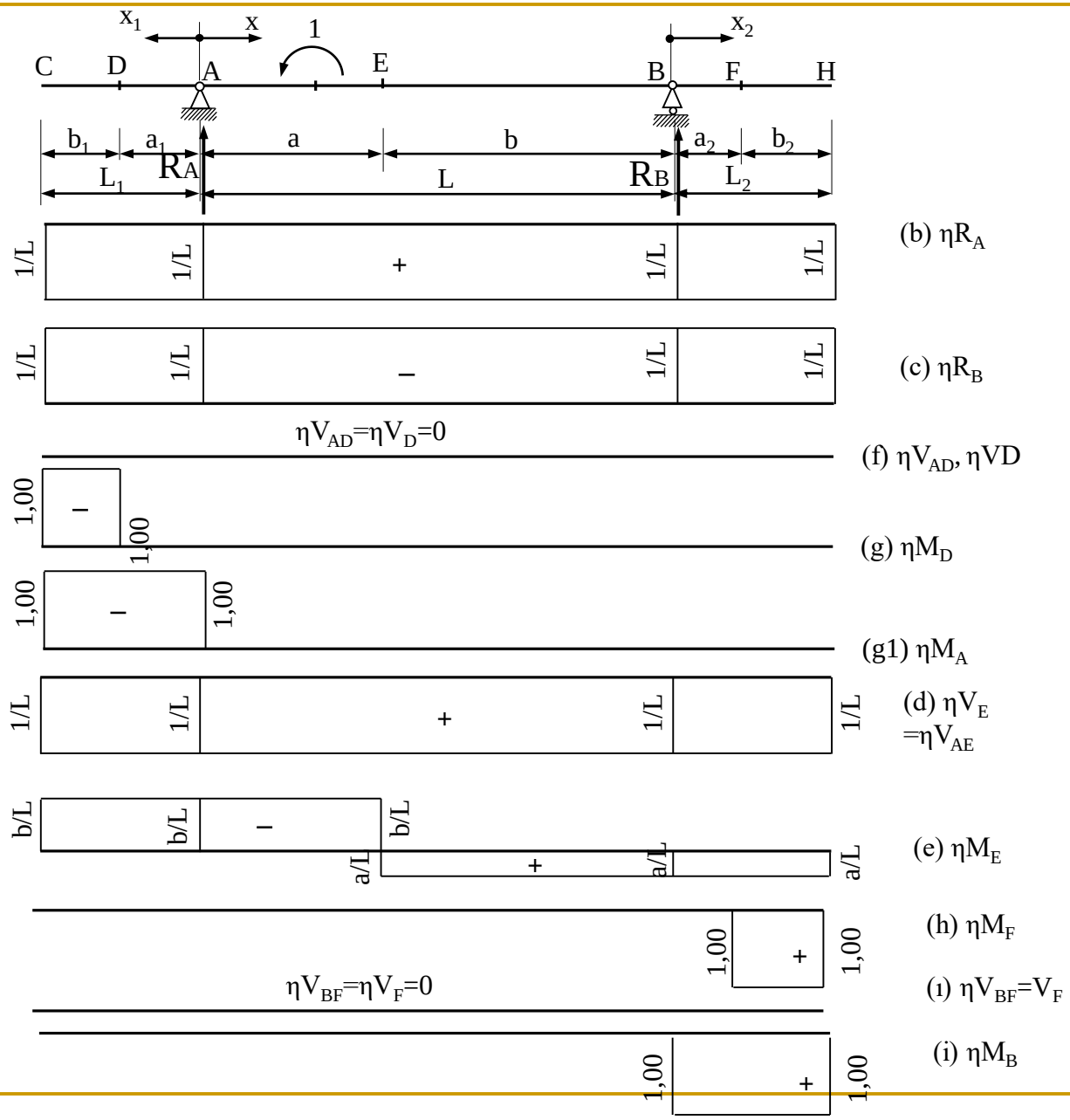
$0<x_2<a_2$

- $V_F = 0$; $\eta V_F = 0$
- $M_F = 0$; $\eta M_F = 0$

$a_2<x_2<L_2$

- $V_F = 0$; $\eta V_F = 0$
- $M_F = 1$; $\eta M_F = 1$




x1
x
1
x2
C
D
A
E
B
F
H
b1
a1
a
b
a2
b2
L1
RA
L
RB
L2
1/L
+
–
(b) ηRA
(c) ηRB
ηVAD=ηVD=0
(f) ηVAD, ηVD
1,00
(g) ηMD
(g1) ηMA
(d) ηVE
=ηVAE
b/L
a/L
(e) ηME
(h) ηMF
ηVBF=ηVF=0
(ı) ηVBF=VF
(i) ηMB

Reference:

All the figures have been taken from the book given below:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ